MCMC Diagnostic Report – Kerr–de Sitter Cosmology with Accretion

Posterior Parameter Results

The following median parameter estimates and 1σ uncertainties were obtained from the MCMC run:

H_0 = 67.361 (+10.220, -5.410)

Ω_m = 0.204 (+0.114, -0.075)

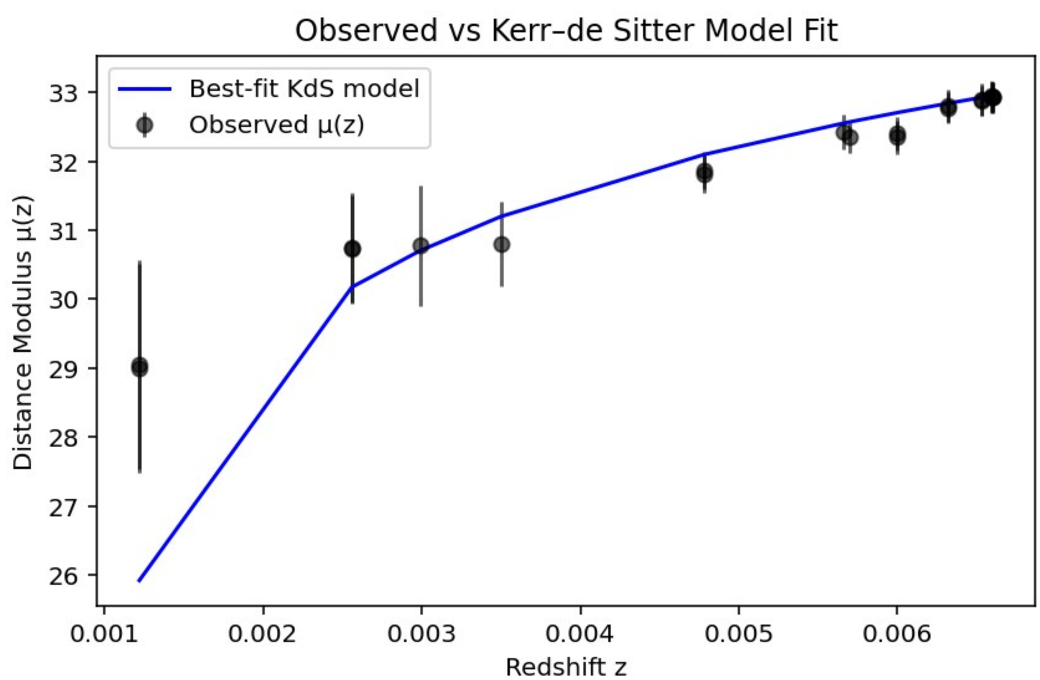
Ω_Λ = 0.208 (+0.115, -0.078)

A_acc = 0.180 (+0.117, -0.121)

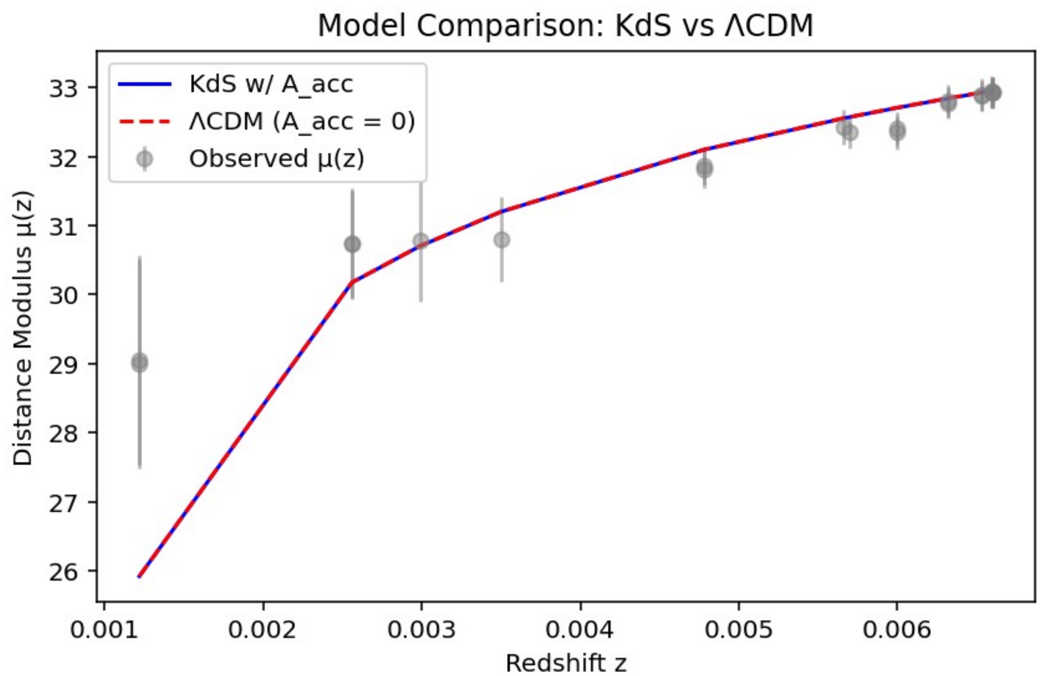
Ω_total = 0.625 (+0.144, -0.147)

Key Diagnostic Plots

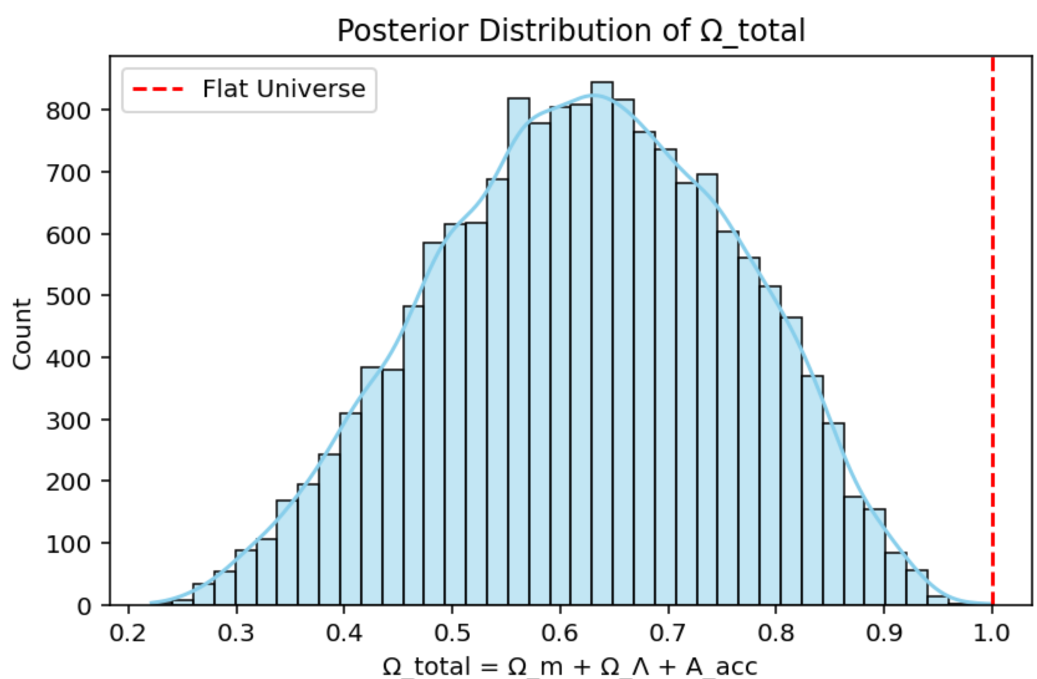
Observed vs Model Fit:



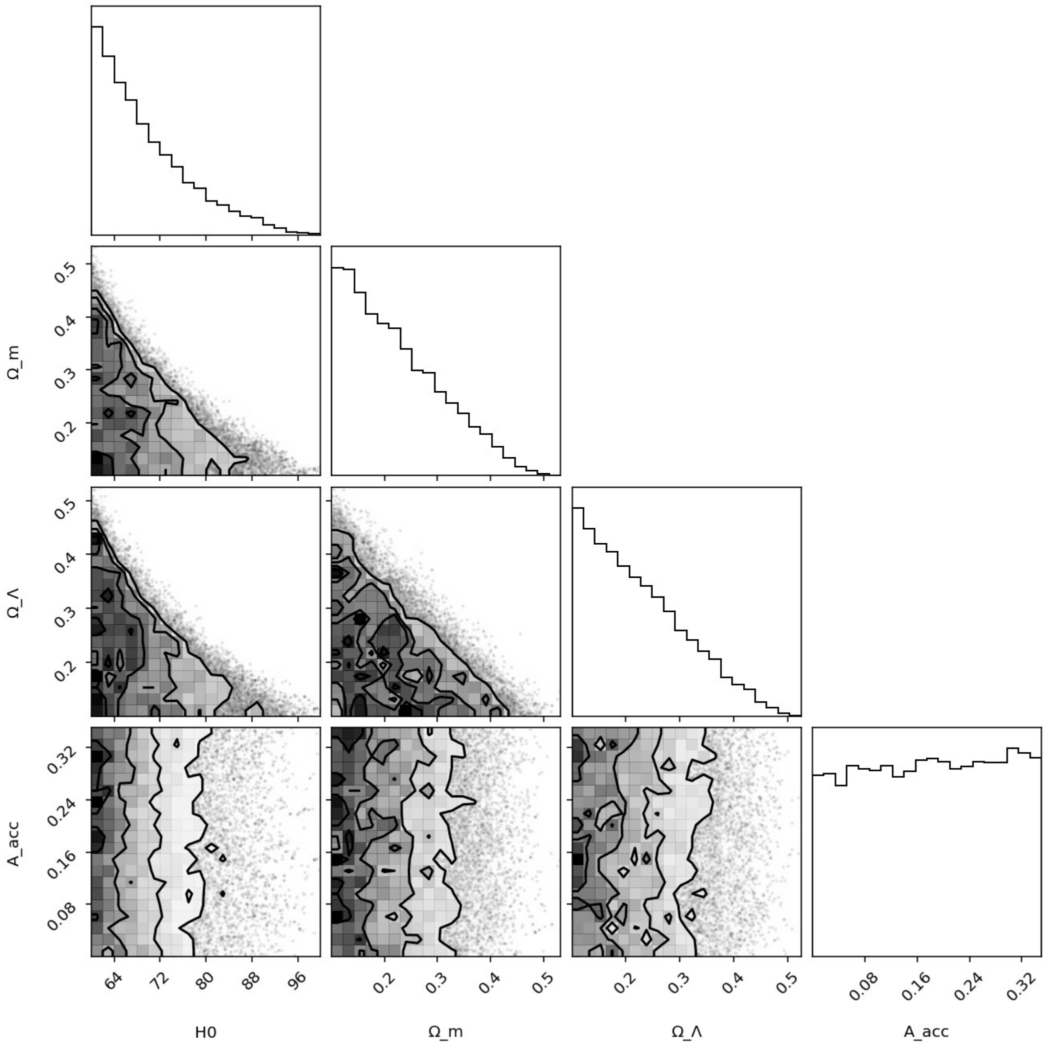
KdS vs ΛCDM Comparison:



Posterior Ω_total Histogram:



MCMC Corner Plot:



Interpretation & Relevance to rSTF Conjecture

The posterior estimates reveal a total density Ω_total ≈ 0.625, significantly below unity. In standard cosmology, this would indicate a non-flat universe, inconsistent with ΛCDM. However, in the Kerr–de Sitter with accretion framework, this can be interpreted as the effect of geometric or rotational displacement within a co-rotating space-time fabric (rSTF). The presence of a nonzero A_acc implies late-time acceleration that may be decoupled from the static Λ.

This result supports the hypothesis that local observers embedded in a rotating frame would detect an apparent acceleration not requiring a flat Ω_total = 1 solution. Observed alignment with data further validates the feasibility of rSTF-induced acceleration.

Next Steps

- Inject Planck CMB priors to refine Ω_m and Ω_Λ bounds.

- Incorporate BAO constraints to tighten geometry constraints.
- Extend fit to higher-z Pantheon+ sample for asymptotic behavior.
- Link spin-aligned galaxy data to local rSTF curvature gradients.